fn approximate_to_tradeoff

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This proof resides in **"contrib"** because it has not completed the vetting process.

1 Hoare Triple

Precondition

Compiler-verified

- Argument `epsilon` of type `f64`
- Argument `delta` of type `f64`

Caller-verified

`epsilon` is finite and nonnegative, and `delta` is in $[0, 1]$.

Pseudocode

```
def approximate_to_tradeoff(
    epsilon: f64,
    delta: f64,
) -> Callable[[RBig], RBig]:
    epsilon = FBig.try_from(epsilon)
    delta = RBig.try_from(delta)

    precision = epsilon.precision().max(10)
    epsilon = epsilon.with_precision(precision).value()

    exp_eps = RBig.try_from(epsilon.with_rounding().exp())  #
    exp_neg_eps = RBig.try_from((-epsilon).with_rounding().exp())  #

    def tradeoff(alpha: RBig) -> RBig:  #
        t1 = RBig(1) - delta - exp_eps * alpha
        base = max(RBig(1) - delta - alpha, RBig(0))
        t2 = exp_neg_eps * base
        return max(max(t1, t2), RBig(0))

    return tradeoff
```

Postcondition

Theorem 1.1. The pseudocode returns a function $\tilde{f}$ that is conservative with respect to the intent I. For `Reporting`, for all $\alpha \in [0, 1]$,

$$\tilde{f}(\alpha) \le f_{\epsilon,\delta}(\alpha).$$

For `Calibration`, for all $\alpha \in [0,1]$,

$$\tilde{f}(\alpha) \geq f_{\epsilon,\delta}(\alpha).$$

Here $f_{\epsilon,\delta}$ is the exact approximate-DP tradeoff curve.

We start with the following definition of DP from Dong, Roth, and Su 2022.

Definition 1.2 (Dong, Roth, and Su 2022, Proposition 2.5)**.** Let $\epsilon > 0$ and $\delta \geq 0$, and define

$$f_{\epsilon,\delta}(\alpha) = \max(0, 1 - \delta - \exp(\epsilon)\alpha, \exp(-\epsilon)(1 - \delta - \alpha)). \tag{1}$$

Then we say that a mechanism M satisfies (ϵ, δ)-DP if it satisfies $f_{\epsilon,\delta}$-DP.

Proof. The definition assumes $\alpha \in [0,1]$, which is the domain of the returned function. The code clamps the base of the second affine branch at zero, as in Definition 1.2.

For `Reporting`, `Dir[I]` is upward rounding and `Opp[I]` is downward rounding. Thus line 11 over-estimates $\exp(\epsilon)$, and line 12 under-estimates $\exp(-\epsilon)$. The results of these constants are converted exactly into rationals to be used in the tradeoff function.

The function defined on line 14 implements the formula in Definition 1.2 with exact fractional arithmetic, except that it substitutes these conservative rounded constants for $\exp(\epsilon)$ and $\exp(-\epsilon)$.

For $\alpha \in [0,1]$, the first branch

$$1 - \delta - \texttt{exp\_eps} \cdot \alpha$$

is no greater than

$$1 - \delta - \exp(\epsilon)\alpha,$$

because $\texttt{exp\_eps} \geq \exp(\epsilon)$ and $\alpha \geq 0$.

For the second branch, if $1 - \delta - \alpha \geq 0$, then

$$\texttt{exp\_neg\_eps} \cdot (1 - \delta - \alpha) \leq \exp(-\epsilon)(1 - \delta - \alpha),$$

because $\texttt{exp\_neg\_eps} \leq \exp(-\epsilon)$. If instead $1 - \delta - \alpha < 0$, then both expressions are negative, so they do not affect the outer maximum with 0.

Therefore, for all $\alpha \in [0,1]$, the function returned by the pseudocode is pointwise no greater than $f_{\epsilon,\delta}(\alpha)$. This is the desired reporting behavior because smaller β means a weaker reported tradeoff curve.

For `Calibration`, the same monotonicity argument reverses. Now `Dir[I]` rounds downward and `Opp[I]` rounds upward, so line 11 under-estimates $\exp(\epsilon)$ and line 12 over-estimates $\exp(-\epsilon)$. The first branch is therefore no smaller than the exact first branch, and the second branch is no smaller than the exact second branch whenever the nonnegative base contributes. Taking the maximum with zero preserves this pointwise inequality, so $\tilde{f}(\alpha) \geq f_{\epsilon,\delta}(\alpha)$. This is the desired calibration behavior because larger β means a stronger tradeoff curve. ☐

References

Dong, Jinshuo, Aaron Roth, and Weijie J. Su (2022). "Gaussian Differential Privacy". In: *Journal of the Royal Statistical Society Series B: Statistical Methodology* 84.1, pp. 3–37. ISSN: 1369-7412.